Pablo García Aznar

PhD · Sustainable Chemistry

Instituto de Tecnología Química, Universitat Politècnica de València, Av. dels Tarongers s/n, 46022 València, Valencia, Spain

pgarazn@itq.upv.es | orcid.org/0000-0003-0586-0037 | pgarazn-chem | Researchgate Profile

Education

Universitat Politècnica de València *València, Spain*

PhD in Sustainable Chemistry *12/2022 - 09/2026*

- Advisor: Dr. Germán Sastre
- PhD Thesis: Computational study and design of graphene-based and MXene-based catalysts for chemical, energy, and environmental applications.

Universitat de València *Burjassot, València, Spain*

MSc in Organic Chemistry *09/2021 - 07/2022*

- Advisor: Dr. Jorge Escorihuela
- Master's Thesis: Computational study on the inverse electron-demand Diels-Alder reaction of norbornenes with 1,2,4,5-tetrazines.

Universitat de València *Burjassot, València, Spain*

BSc in Chemistry *09/2017 - 07/2021*

- Undergrad research advisor: Dr. Jorge Escorihuela
- Bachelor's thesis: Synthesis of chiral homopropargylic amides for intermolecular Pauson-Khand reactions.

Professional Experience

12/2022 - 10/2026	**PhD Research contract**, Theoretical Chemistry Department, Instituto de Tecnología Química (UPV-CSIC), Universitat Politècnica de València, Valencia, Spain.
11/2021 - 07/2022	**Laboratory Intern**, Organic Chemistry department, University of Valencia, Burjassot, Valencia, Spain.

Expertise

Reaction Mechanisms via DFT: Extensive experience mapping complex reaction pathways and identifying transition states for both homogeneous organic systems and complex catalytic environments. Proficient in electronic structure codes (Gaussian, ORCA, CASTEP) to elucidate reaction mechanisms and calculate energy barriers.

Machine-Learning Potentials (MLIPs): Experienced in leveraging machine learning interatomic potentials (MACE) within ASE to evaluate structural stability and run dynamic simulations. Skilled at generating robust DFT reference datasets to expand these potentials into new reactive regimes.

Proficient in Python and ASE, including developing custom automated workflows (*e.g.*, AutoNEB) for transition state searches. Capable of adapting these high-throughput computational workflows to conduct in silico library prescreening or evaluate multiple active site variants. Experienced in handling large-scale parallel calculations.

Theoretical-Experimental Collaboration: Proven track record of working alongside experimental teams to bridge theory and laboratory practice. Responsible for building computational models that are accurately based on experimental data, such as XPS spectra, and provide mechanistic rationale for measured macroscopic reactivity.

Local LLM deployment and setup: Experience with local LLM evaluation and deployment on both local hardware and API serving for agentic tasks, experience with AI harnesses (OpenCode, Cline).

Wet Lab Experience: Organic synthesis, standard laboratory operations, and structural characterization (specifically IR, ^{1}H and ^{13}C NMR).

Publications

Proven track record of translating quantum chemical outputs into mechanistic descriptions for experimental synthesis teams. In the co-authored experimental works listed below, I served as the lead computational researcher, responsible for the original model development, reaction mechanism elucidation, and simulation frameworks.

Published - Lead computational researcher

Andrés Mollar-Cuni, **Pablo García-Aznar**, Santiago Martín, German Sastre, Hermenegildo García, Jose A. Mata. Mechanistic Insights in Acceptorless Dehydrogenation of N-Heterocycles Using Graphenes as Carbocatalysts. *ACS Appl. Mater. Interfaces*, **2025**, 17, 51, 69234–69244, doi: https://doi.org/10.1021/acsami.5c11895

Aicha Anouar, Elena Romero Salicio, **Pablo García-Aznar**, Abdessamad Grirrane, Amparo Forneli, German Sastre, Hermenegildo Garcia, Ana Primo. Ti_3C_2 MXene as heterogeneous catalyst for carbodiimides guanylation with over 10^2 h^{-1} turnover frequencies. *J. Catal.*, 450, **2025**, 116328, doi: https://doi.org/10.1016/j.jcat.2025.116328

María Cabrero-Antonino, Andrés Uscategui-Linares, Rubén Ramírez-Grau, **Pablo García-Aznar**, German Sastre, Jianjun Zhang, Sara Goberna-Ferrón, Josep Albero, Jiaguo Yu, Hermenegildo García, Feiyan Xu, Ana Primo. 2D/2D MOF/MXene Schottky Junction: Prolonged Carrier Lifetime and Enhanced Hydrogen Evolution Efficiency. *Angew. Chem. Int. Ed.* **2025**, 64, e202503860, doi: https://doi.org/10.1002/anie.202503860

Rubén Ramírez Grau, **Pablo García-Aznar**, German Sastre, Sara Goberna-Ferrón, Octavian Pavel, Alina Tirsoaga, Bogdan Cojocaru, Dana Georgeta Popescu, Vasile I. Parvulescu, Ana Primo, and Hermenegildo García. MXenes as Heterogeneous Thermal Catalysts: Regioselective Anti-Markovnikov Hydroamination of Terminal Alkynes with 10^2 h^{-1} Turnover Frequencies. *J. Am. Chem. Soc.* **2025**, 147, 4, 3315–3332, doi: https://doi.org/10.1021/jacs.4c13481

Xuechao Tan, **Pablo García-Aznar**, German Sastre, Suk Bong Hong. Hydrothermal Aging Enhances Nitrogen Oxide Reduction over Iron-Exchanged Zeolites at 150 °C. *J. Am. Chem. Soc.* **2024**, 146, 9, 6352–6359, doi: https://doi.org/10.1021/jacs.4c00248

Yingying Jiao, Liang Tian, **Pablo García-Aznar**, German Sastre, Yike Li, Jianshe Wang, Zhanhang He, Zhongjun Li, Hermenegildo García. Enhancement of intramolecular charge transfer in carbon nitride by attaching 2,4,5-Trichlorophenoxyacetic acid as electron acceptor units. *Chem. Eng. J.*, **2023**, 473, 145248, doi : https://doi.org/10.1016/j.cej.2023.145248

Pablo García-Aznar, Jorge Escorihuela. Computational insights into the inverse electron-demand Diels–Alder reaction of norbornenes with 1,2,4,5-tetrazines: norbornene substituents' effects on the reaction rate. *Org. Biomol. Chem.*, **2022**, 20, 6400-6412, doi: https://doi.org/10.1039/D2OB01121B

Co-authored

Octavian Pavel, Alina Tirsoaga, Bogdan Cojocaru, Dana Popescu, Ruben Ramírez-Grau, Pablo González-Durán, **Pablo García-Aznar**, Liang Tian, German Sastre, Ana Primo, Vasile Parvulescu and Hermenegildo Garcia. Nb_2C Mxene as a bifunctional acid–base and oxidation/hydrogenation catalyst. *EES Catal.*, **2025**,3, 856-869, doi: https://doi.org/10.1039/D5EY00004A

Alechania Misturini, Omer F. Altundal, **Pablo García-Aznar**, Sara Kariminasab, German Sastre. Effect of Intracrystalline Silanol Defects on the Diffusivity of Benzene in Silicalite Zeolite *Chem. Ing. Tech*, **2023**, 95: 1768-1776, doi: https://doi.org/10.1002/cite.202300008

In Preparation

Mechanistic insight into Electrochemical Hydrogen Evolution on Two-Dimensional MXenes: A Computational screening Study

Effectivity of surface double vacancies in MXenes for CO_2 hydrogenation to formic acid using big data and a machine-learning force field.

Presentations

Organized events

Jun 2024. Part of the organizing committee of the EUROMXENE 2024 Congress, held at the Polytechnic University of Valencia (UPV) in Valencia, Spain (Link to the website).

Contributed Presentations

Presentations

Pablo García-Aznar, Andrés Mollar-Cuni, Santiago Martín, German Sastre, Hermenegildo García, Jose A. Mata. February 2026. Mechanistic Insights in Acceptorless Dehydrogenation of N-Heterocycles Using Graphenes as Carbocatalysts. 8th Winter Meeting 2026, Valencia, Spain

Pablo García-Aznar, Germán Sastre, Rubén Ramírez Grau, Ana Primo, Hermenegildo García. February 2024. MXenes as heterogeneous catalysts. Computational insight into the regioselective anti-Markonikov hydroamination of terminal alkynes. XV Conference of Young Researchers in Atomic and Molecular Physics, in ICMol, Burjassot, València, Spain, (Link to the abstract; page 60)

Pablo García-Aznar, Germán Sastre, Rubén Ramírez Grau, Ana Primo, Hermenegildo García. September 2023. MXenes as heterogeneous catalysts. Computational insight into the regioselective anti-Markonikov hydroamination of terminal alkynes. ITQ - Catalysis Hub meeting at the *Diamond Light Source* in Abingdon, United Kingdom.

Poster presentations

Pablo García-Aznar, Germán Sastre, Aicha Anouar, Elena Romero Salicio, Abdessamad Grirrane, Amparo Fornelli, Hermenegildo García, Ana Primo. July 2025. Computational insight into the guanylation of carbodiimides via $Ti_3C_2T_x$ MXene as a heterogeneous catalyst. Poster: CCP5 Summer School 2025, Newcastle, UK .

Pablo García-Aznar, Germán Sastre, Aicha Anouar, Elena Romero Salicio, Abdessamad Grirrane, Amparo Fornelli, Hermenegildo García, Ana Primo. February 2025. Computational insight into the guanylation of carbodiimides via $Ti_3C_2T_x$ MXene as a heterogeneous catalyst. Poster: 7th Winter Meeting 2025, Valencia, Spain .

Pablo García-Aznar, Germán Sastre, Aicha Anouar, Elena Romero Salicio, Abdessamad Grirrane, Amparo Fornelli, Hermenegildo García, Ana Primo. June 2024. Computational insight into the guanylation of carbodiimides via $Ti_3C_2T_x$ MXene as a heterogeneous catalyst. Poster: EUROMXENE Congress 2024, Valencia, Spain .

Pablo García-Aznar, Yingying Jiao, Liang Tian, German Sastre, Yike Li, Jianshe Wang, Zhanhang He, Zhongjun Li, Hermenegildo García. September 2023. Enhancement of intramolecular charge transfer in carbon nitride by attaching 2,4,5 Trichlorophenoxyacetic acid as electron acceptor units. Poster: Workshop "Tailoring Reaction Pathways by Catalysis", Liblice, Czechia.

Several poster presentations at the annual Doctoral Student Meeting at the Polytechnic University of Valencia, Valencia, Spain, *25 May 2023 (VIII), 23 May 2024 (IX), 22 May 2025 (X)*. Link 2023, Link 2024, Link 2025 .

Pablo García-Aznar, Jorge Escorihuela. May 2022. Computational investigations on the reactivity of inverse electron-demand Diels-Alder reactions of fused-norbornenes with 3,6-dipyridin-2-yl-1,2,4,5-tetrazine. Poster: 2nd Workshop for Young Researchers in Chemistry, Burjassot, Valencia, Spain (Link to abstract, page 51).

Outreach & Professional Development

March 27, April 10 and May 8, 2025 **CiènciaLAB (scientific outreach workshop)**, Volunteer — *Universitat Politècnica de València*

Additional Training

Data Analysis using AI tools, *November-December 2024 (20h), in Universitat Politècnica de València*, where I learned basic concepts in data science and how to interpret and extract information from big data (using RapidMiner).

CCP5 Summer School 2025, *20th - 31st July 2025, in Newcastle University*. I learned basic Python skills and notions in many computational chemistry modelling techniques, mainly molecular dynamics and Monte Carlo methods. I also learned basic use of ASE and MACE Machine-Learning interatomic potentials, simple MLIP fine tuning and testing.

Digital Competencies for Comp. Biology, *February - April 2026, Online*. I advanced Python skills and learned Nextflow and the uv toolchain to ensure high-throughput automation and computational reproducibility.

Peer Review

Reviewer for RSC advances, 3 reviewed articles.

Languages

Spanish (Native), Catalan (C1), English (C1), French (A2)